Started on	Tuesday, 8 November 2022, 9:54 AM
State	Finished
Completed on	Tuesday, 8 November 2022, 9:59 AM
Time taken	4 mins 59 secs
Grade	**8.00** out of 10.00 (**80**%)

Question **1**

Correct

Mark 2.00 out of 2.00

How can we pass multiple flags to system calls?

Select one:

- a. Using the bitwise-or operator `|`. ✔
- b. Chaining all flags with `,` in the function.
- c. Using the bitwise-and operator `&`.
- d. There is a symbol for every combination of flags.

The flag argument is a number.
The operating system checks if each bit in the flag argument is set.
To set bits in a number we use the bitwise-or operator.

The correct answer is: Using the bitwise-or operator `|`.

Question **2**

Correct

Mark 2.00 out of 2.00

How can we allocate memory in the .stack section?

Select one:

- a. Declare a variable in a function. ✔
- b. Use the `malloc` function family.
- c. Use the `static` keyword.
- d. The stack is only modified by the OS, so we can't influence it.

All variables we declare and use inside functions, without any additional type keywords, are put in the .stack memory zone.

The correct answer is: Declare a variable in a function.

Question **3**

Correct

Mark 2.00 out of 2.00

The operating system's kernel does NOT:

Select one:

- a. Update system packages. ✔
- b. Manage access to hardware resources.
- c. Isolate processes.
- d. Provide a hardware abstraction layer.

The operating system manages system resources and provides a common interface to userspace applications to improve the system's security and application compatibility. Updating system packages is one of the tasks that system administrators must perform either manually or using automatic package managers.

The correct answer is: Update system packages.

Question **4**

Correct

Mark 2.00 out of 2.00

Which loads faster between a dynamic and a static executable?

Select one:

- a. A dynamic executable ✔
- b. A static executable
- c. They load in the same time frame.
- d. Loading does not depend on the executable type.

Dynamic executables have a smaller size, and, as such, are loaded faster.
Static executables include all the library code, which needs to be brought into memory first.

The correct answer is: A dynamic executable

Question **5**

Incorrect

Mark 0.00 out of 2.00

Why can we sometimes run code saved in the `.rodata` region?

Select one:

- a. The zone is coallesced in the `.text` region.
- b. The zone is read-only, so we can't execute code. ✖
- c. The zone can be forced to have execution rights at start.
- d. The `.rodata` zone was randomly given execution permissions during the run.

.If the `.rodata` zone is small, it can sometimes be merged with the `.text` region by the compiler.
This behaviour is not consistent and should not be depended on.

The correct answer is: The zone is coallesced in the `.text` region.

◄ Curs 04: Date: Recap

Jump to...

Curs 06: Calcul (partea a 2-a) ►